

Code File: Converting DICOM Images to PNG/JPG

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"""
AI-CliniScan: DICOM to PNG Conversion and Image Preprocessing
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Description: Extracts pixel arrays from DICOM chest X-rays, handles MONOCHROME inversion,
normalizes intensities, resizes, and applies CLAHE contrast enhancement.
"""

import os
import cv2
import pydicom
import numpy as np
from glob import glob
from tqdm import tqdm

# Configuration Paths
INPUT_DIR = "/kaggle/input/vinbigdata-chest-xray-abnormalities-detection/train"
OUTPUT_DIR = "/kaggle/working/processed_train"
os.makedirs(OUTPUT_DIR, exist_ok=True)
TARGET_SIZE = (512, 512)

def process_dicom_to_png(dicom_path, save_path):
    try:
        # 1. Read DICOM file
        dicom = pydicom.dcmread(dicom_path)
        pixel_array = dicom.pixel_array

        # 2. Fix Monochrome (Invert if MONOCHROME1 so bones are white)
        if dicom.PhotometricInterpretation == "MONOCHROME1":
            pixel_array = np.max(pixel_array) - pixel_array

        # 3. Min-Max Intensity Normalization (0-255)
        pixel_array = pixel_array.astype(np.float32)
        pixel_array = (pixel_array - np.min(pixel_array)) / (np.max(pixel_array) - np.min(pixel_array)) * 255.0
        pixel_array = pixel_array.astype(np.uint8)

        # 4. Resize to uniform dimensions
        img_resized = cv2.resize(pixel_array, TARGET_SIZE)

        # 5. Apply CLAHE (Contrast Limited Adaptive Histogram Equalization)
        clahe = cv2.createCLAHE(clipLimit=2.0, tileGridSize=(8,8))
        img_final = clahe.apply(img_resized)

        # Save as PNG
        cv2.imwrite(save_path, img_final)
        return True

    except Exception as e:
        print(f"Error processing {dicom_path}: {e}")
        return False

if __name__ == "__main__":
    dicom_files = glob(f"{INPUT_DIR}/*.dicom")
    print(f"Starting conversion for {len(dicom_files)} images...")

    for file_path in tqdm(dicom_files):
        file_name = os.path.basename(file_path).replace('.dicom', '.png')
        save_path = os.path.join(OUTPUT_DIR, file_name)
        process_dicom_to_png(file_path, save_path)

    print("DICOM to PNG conversion complete.")
```